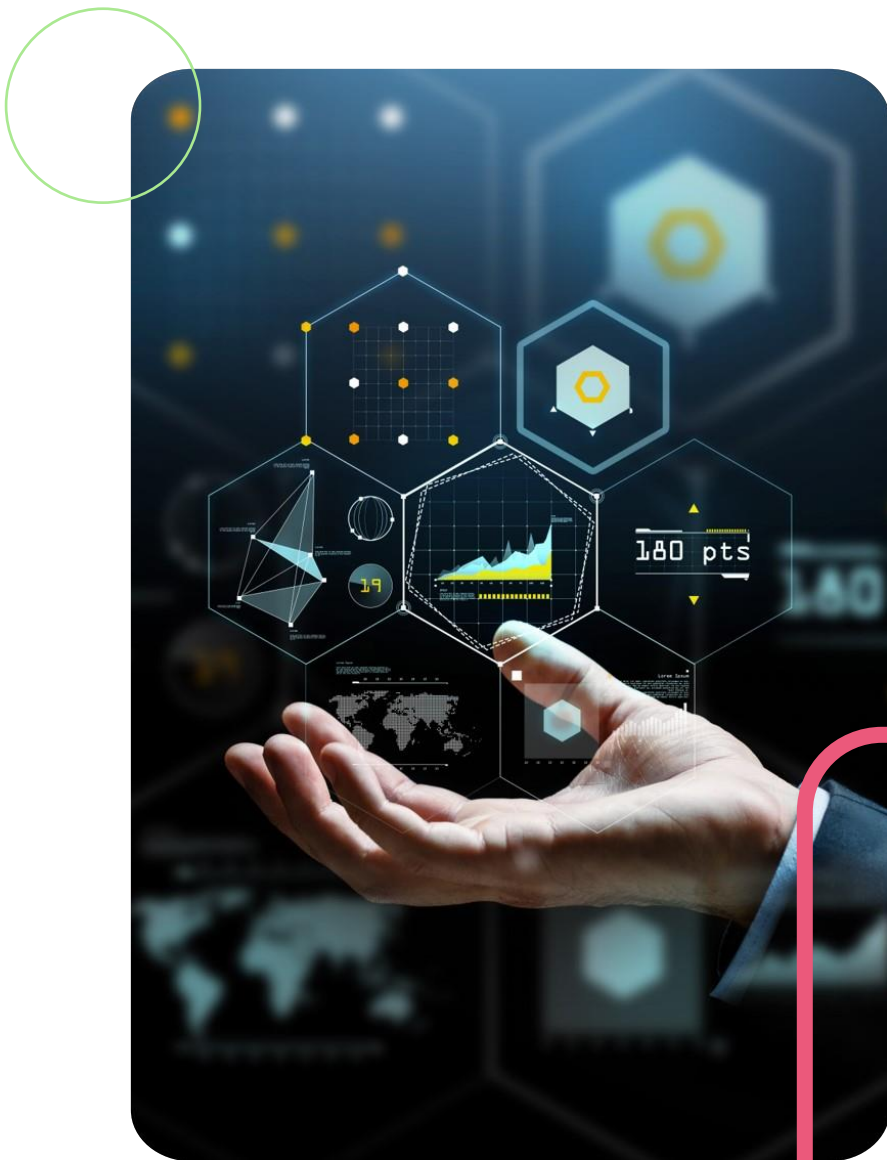




Diploma in **Data Analytics**

Lesson 5: Dates and times



Dates and Times continued <

Dates and times data wrangling essentials <

Time series analysis intro <

Lesson Objectives

Lesson 5

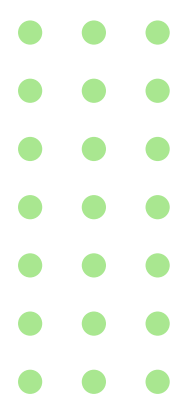


Dates & times continued...

Recap previous lesson

- `Sys.timezone()`
- `Sys.date()`
- `Sys.time()`
- `Now()`
- `As.date()`
- `Ymd()` or `mdy()`





Did
you
know?

1 January 1970

R start date

Compares all dates to origin date with logical comparisons.



Lubridate

- Tidyverse package
 - Integrates well with other Tidyverse packages
- Package that simplifies date and time handling
- Treats underlying date objects the same



3 types of date/time data in R

- Date
 - `<date>`
- Time
 - `<time>`
- Date-time
 - `<dtm>`

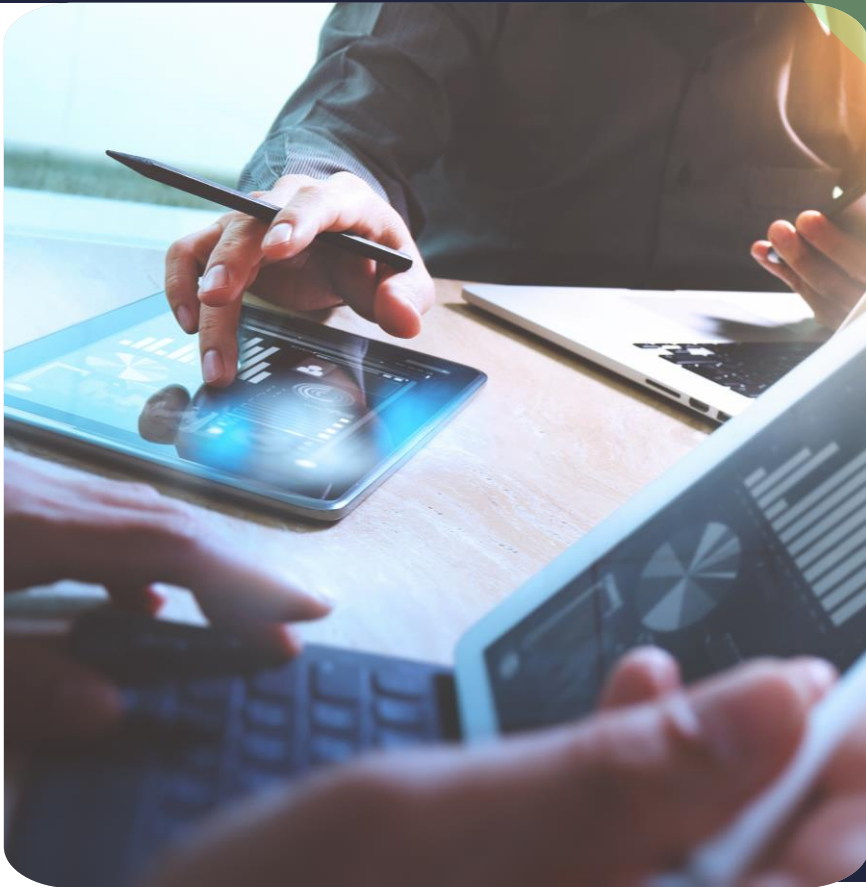


More lubridate functionality

Parsing dates

`ymd()`, `mdy()`, `dmy()`

`ymd_hms()`, `ymd_hm()`, `ymd_ms()`

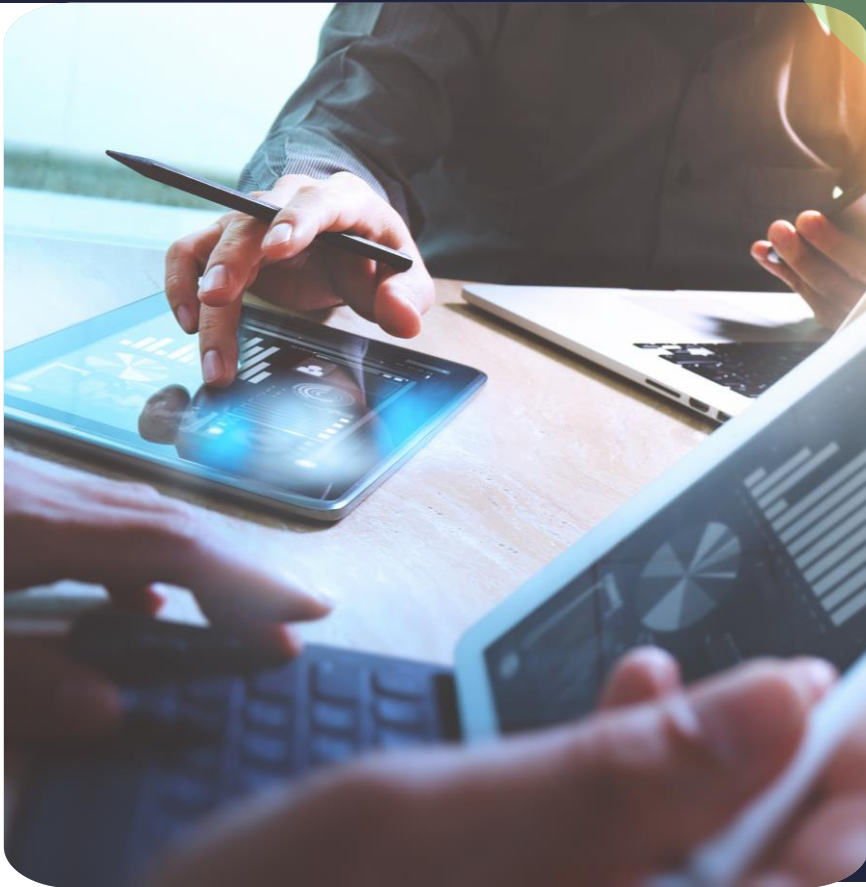


More lubridate functionality

Time duration

`duration()`, `is.duration()`, `as.duration()`

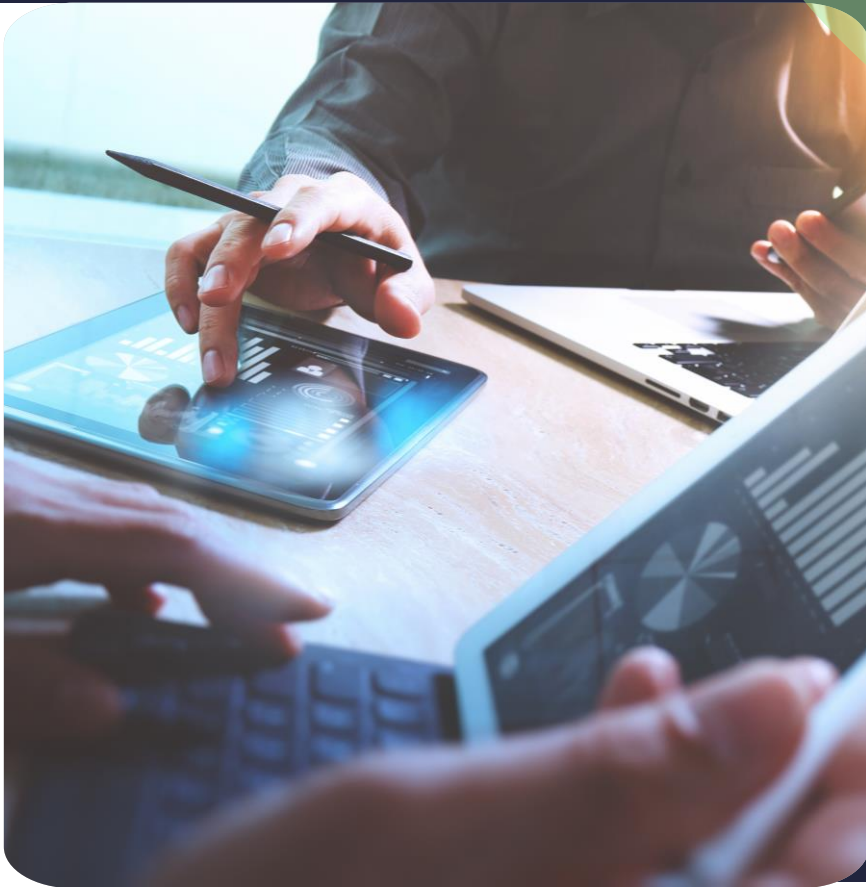
`dseconds()`, `dminutes()`, `dhours()`, `ddays()`,
`dweeks()`, `dyears()`

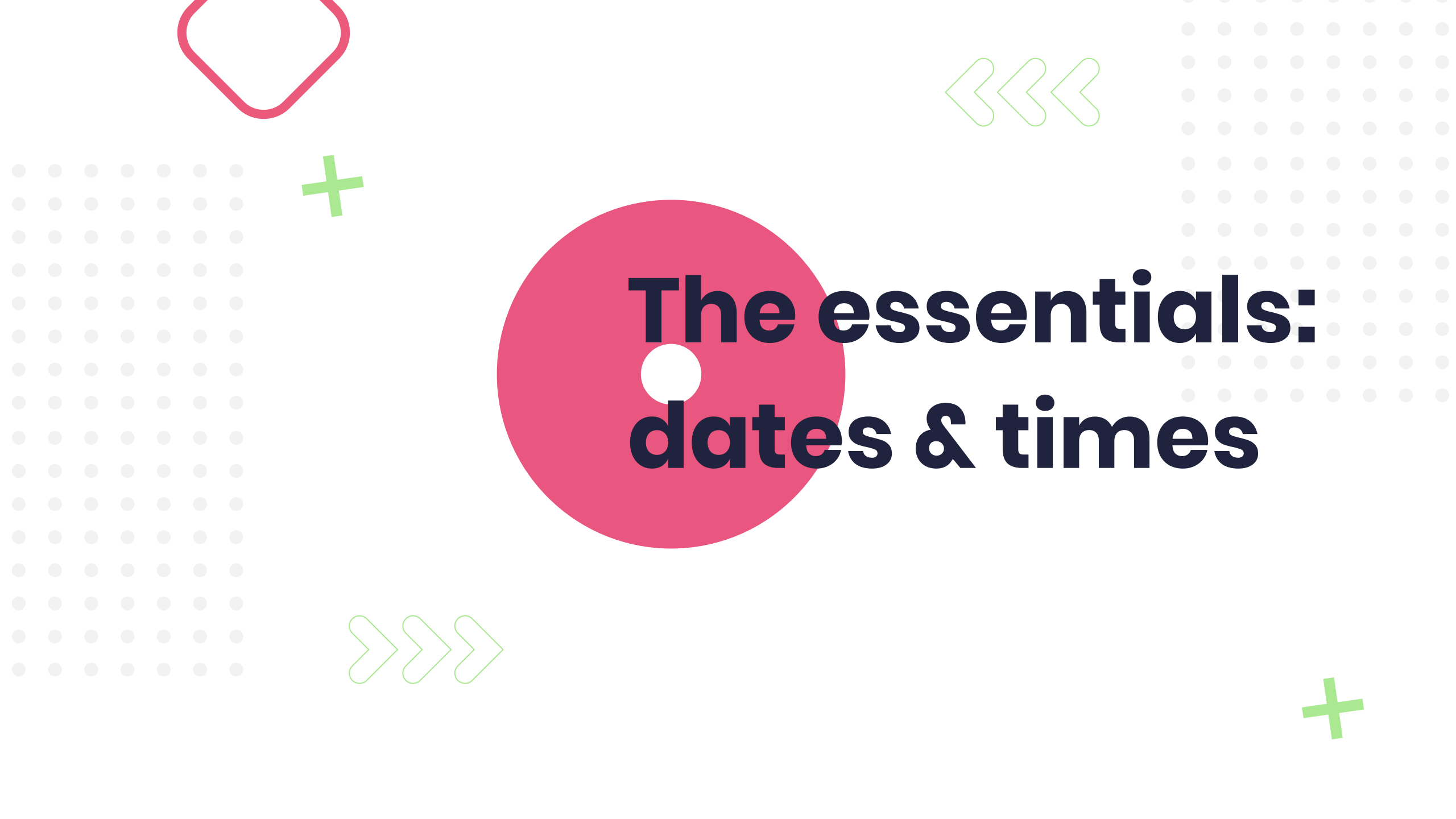


More lubridate functionality

Setting and extracting information

`second()`, `minute()`, `hour()`, `day()`, `wday()`,
`yday()`, `week()`, `month()`, `year()`, `tz()`





The essentials: dates & times

Recap

- 3 types (date, time, date-time)
- today(), now()
- Ymd(), mdy(), dmy(), -_hms(), and -_hm()



Essential lubridate functionality

Create a date/time from individual items

Make_date() and make_datetime()



Essential lubridate functionality

Rounding dates

ceiling_date(), floor_date(), and
round_date()



Essential lubridate functionality

Intervals

is.interval(), as.interval(), interval(),
int_shift(), int_flip(), int_aligns(),
int_overlaps(), and %within%



Essential lubridate functionality

Setting components

`datetime()`

`update()`

<https://cran.r-project.org/web/packages/lubridate/lubridate.pdf>





Time series analysis intro

Time series

- A variable that is measured over time in sequential order
- Ex. Weekly production output, monthly sales, annual rainfall
- Analysis of time series leads to identifying trends that explain variations



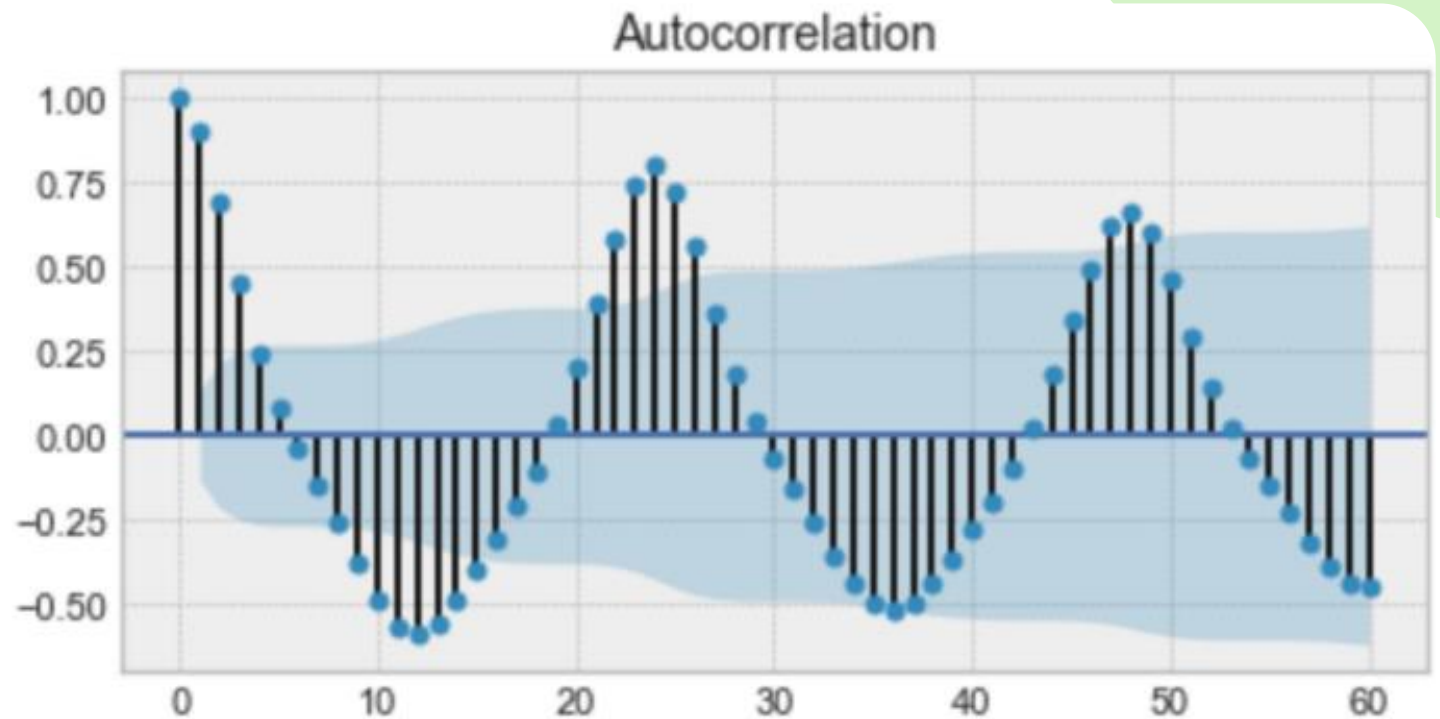
Why time series

- Help predict trends over time
- Evaluate if trend is seasonal, stationary or if target variable is autocorrelated



Autocorrelation

Degree of similarity between observations as a function of the time lag amongst them



Source: <https://towardsdatascience.com/almost-everything-you-need-to-know-about-time-series-860241bdc578>

Seasonality

The existence of variations that take place at specific methodical intervals less than a year apart, for instance weekly, monthly or quarterly.



Source: <https://towardsdatascience.com/almost-everything-you-need-to-know-about-time-series-860241bdc578>

Stationarity

Time series is assumed to be stationary if the mean, variance and covariance does not change over time.



Source: <https://towardsdatascience.com/almost-everything-you-need-to-know-about-time-series-860241bdc578>

Regression techniques

- Moving average
- Seasonal analysis
- Exponential smoothing
- ARIMA





Challenge >>

- Download dataset containing dates and time
- Clean and extract information using lubridate package

#exploredata